

Name: _____

Roll Number: _____

Quiz-4**Max. Time: 20 min****Max. Points: 20**

Note: Solve all parts. Limit your written responses to the provided space.

Q.1. [8] Choose by putting a check mark on the most appropriate option. Note: No cutting/overwriting is allowed.i. For a matrix $A_{m \times n}$, $\text{Col } A$ will span $\mathbb{R}^m$ if and only if A maps $\mathbb{R}^n$ _____ $\mathbb{R}^m$

(A) one-to-one (B) onto (C) Both 'A' and 'B' (D) None of the given options

ii. A vector space is not a subspace.

(A) True (B) False

iii. Any line through origin in $\mathbb{R}^3$ is a 3-dimensional subspace.

(A) True (B) False

iv. A set of vectors bounded by a circle centered at the origin in $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$.

(A) True (B) False

v. The set of points defining the intersection of two planes in $\mathbb{R}^3$ is a vector space if:

(A) First plane passes through the origin (B) Second plane passes through the origin

(C) The line of intersection contains the origin (D) It's always a vector space irrespective of the location of intersection

vi. The following set describes a basis for $\mathbb{R}^3$.

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \right\}$$

(A) True (B) False

vii. A linearly independent set in a subspace H is its basis.

(A) True (B) False

viii. Vector $\mathbf{0}$ is a subspace.

(A) True (B) False

Q.2. [5+5]

a) Find a spanning set for the null space of the following matrix.

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution:s. Row reduce the augmented matrix $[A \quad \mathbf{0}]$ to reduced echelon form and express the basic variables in terms of the free variables:

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{aligned} x_1 - 2x_2 - x_4 + 3x_5 &= 0 \\ x_3 + 2x_4 - 2x_5 &= 0 \\ 0 &= 0 \end{aligned}$$

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The general solution is $x_1 = 2x_2 + x_4 - 3x_5$, $x_3 = -2x_4 + 2x_5$, with x_2 , x_4 , and x_5 free. Next, decompose the vector giving the general solution into a linear combination of vectors where *the weights are the free variables*. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$= x_2 \mathbf{u} + x_4 \mathbf{v} + x_5 \mathbf{w}$$

Every linear combination of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ is an element of $\text{Nul } A$ and vice versa. Thus $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a spanning set for $\text{Nul } A$.

b) Let $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Determine if $\mathbf{w}$ is in $\text{Col } A$.

Solution:

$$[A \quad \mathbf{w}] \sim \begin{bmatrix} 1 & 0 & 1 & -1/2 \\ 0 & 1 & 1/2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

the system is consistent and $\mathbf{w}$ is in $\text{Col } A$.

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